

Shutdown Engineering

Engineering Cycle 1

Separating Expression Duration from Counter-State Persistence

第一轮先讲清楚为什么需要 shutdown，以及为什么最终选择复用 BM3R1 延迟。核心问题是：计数器已经能保存状态，但下游不能因此一直表达。

Design

Design goal: preserve the DNA-state memory required for counting while allowing downstream expression to decline before the active state ends.

Our initial design connected downstream expression directly to a counter bit. This created a timing limitation: expression could continue for as long as that bit remained active. In a multi-bit counter, a high-order bit may remain active across several input cycles, producing a prolonged output rather than a separately controlled expression window.

We therefore considered several possible solutions. State decoding alone could specify when an output was enabled but did not itself provide a shutdown process. An additional duration counter would introduce further circuit complexity, while protein degradation tags alone would not stop continued protein production. Using the natural periods of different bits offered a simpler way to create predefined windows, but tied their timing to the counter's state sequence.

We selected the existing BM3R1-dependent delay for further development. After a PB→LR transition, residual BM3R1 persists before its level declines sufficiently to relieve repression. We proposed using this same signal to control a separate shutdown branch, allowing expression duration to be adjusted without requiring another DNA-state transition.

比较过其他方案，但重点不在罗列方案，而在说明取舍：利用 bit 周期比较简单，却限制了窗口与开始时间的组合；复用 BM3R1 则提供了一个可以进一步调节的关闭过程。

Build

We proposed a shutdown branch comprising a BM3R1-repressed promoter, an anti-output sRNA, and a compatible target region on the output mRNA.

Following PB→LR switching, LR-dependent output transcription begins while residual BM3R1 temporarily represses sRNA production. As BM3R1 declines, the shutdown promoter becomes more active, allowing sRNA to accumulate and suppress new output production.

This branch reads the same BM3R1 signal as the RDF branch but serves a different function. RDF supports subsequent counter reversal when integrase becomes available; the sRNA branch acts on downstream expression without requiring LR to return to PB.

We implemented this architecture in the oscillator-counter model by adding sRNA, target-mRNA, and protein-readout states. The model also included BM3R1 binding to the added shutdown operators, allowing us to examine whether the new branch affected RDF regulation and counter switching.

Build 把两条支路的职责区分清楚。RDF 解除抑制并不等于输出自然关闭，所以需要额外的 sRNA 支路。我们的实际 Build 是把这个结构写成耦合 ODE。

Test

We tested whether the proposed module could satisfy three requirements: preserve an initial output peak, suppress new translation while LR remained active, and maintain correct counter switching.

In the current corrected model, shutdown performance was assessed using new translation flux. Passing conditions required a finite, positive full width at half maximum (FWHM), late-LR translation below 20% of the pulse peak, and peak retention of at least 50% relative to the corresponding no-shutdown control.

At the nominal 50-minute doubling time, translation-flux FWHM decreased from **10.61 h to 6.98 h**. Late-LR translation fell to **17.0% of peak**, while the initial peak retained **98.6%** of the control. With the modeled shutdown operator load included, **5/5 audited cycles** retained exactly one complete reversal.

We then varied sRNA transcription strength and the BM3R1 repression threshold of the shutdown branch. **91 of 169 combinations** met the shutdown criteria, with passing FWHM values of **5.42–7.36 h**.

测试结果要逐项回应设计要求，而不只是说“模型产生了脉冲”。这里既展示后期输出受到抑制，也检查初始峰值和计数器是否保留。参数扫描进一步说明，表达窗口在模型中具有可调性。

Learn

The simulations supported the proposed separation between DNA-state persistence and downstream expression duration. The counter could remain in LR while the sRNA branch reduced new translation, allowing shutdown to occur before a subsequent counter reversal.

However, the model also clarified two limitations. First, the shutdown branch is not completely independent of the counter because its operators bind BM3R1. Second, suppression of new translation does not imply immediate removal of accumulated protein. At baseline, late-LR

stable protein remained at approximately **32.0% of peak**, even though new translation met the shutdown criterion.

We therefore retained the architecture as a candidate local shutdown mechanism and moved to a second question: **which parameter balances and growth conditions allow it to preserve both a useful initial output and effective late suppression?**

这一轮学到的是“稳定状态不必对应持续的新生输出”，而不是“所有输出都已经被彻底清除”。自然引出下一轮：这个机制有多依赖参数，以及什么条件会导致关闭失败。

Engineering Cycle 2

Defining the Conditions for Reliable Shutdown

第二轮不预设“设计很稳健”，而是研究它的适用范围。早期模型给出了积极结果，后续修正则更准确地认识到：sRNA 的供给、消耗和靶 mRNA 的产生必须匹配。

Design

Design goal: identify the operating conditions that support delayed shutdown and translate its failure modes into design constraints.

The first cycle established a feasible shutdown mechanism, but its performance depended on how rapidly sRNA accumulated relative to continuing target-mRNA production.

This became particularly important during model refinement. The early formulation represented sRNA-dependent target-mRNA loss without a corresponding sRNA-consumption term. The corrected model explicitly accounts for sRNA co-degradation during target removal, using the RyhB/sodB system as a literature-based kinetic proxy.

We therefore examined whether sRNA supply remained sufficient under simultaneous parameter variation, while also testing how growth-dependent changes affected the coupled oscillator, counter, and shutdown dynamics.

最初证明“存在关闭机制”之后，进一步检查模型假设；加入 sRNA 共同消耗后，问题就变成了“它能否持续提供足够的抑制能力”。这也解释了为什么不能继续使用旧版 99.2% 来概括现行模型。

Build

We refined the shutdown equations to include consumption of both sRNA and target mRNA during pairing, and separated literature-based kinetic estimates from declared design assumptions.

To examine growth effects, we regenerated oscillator trajectories at doubling times of **40, 50, and 60 min**, updated protein dilution, and simulated the coupled system with a matched no-

shutdown control for each condition.

We also implemented one-at-a-time sensitivity analysis and a **512-point Latin hypercube sample across ten shutdown parameters**. This joint analysis used the nominal counter trajectory and controls matched to target-mRNA transcription rate; full-system coupling and operator-load effects were assessed separately.

Finally, we constructed a **225-point transcription-supply scan** to examine the relationship between effective sRNA production and target-mRNA production.

这里写清楚实际完成的方法。生长分析重新计算了上游输入；联合参数分析研究下游参数组合；供给扫描则专门解释为什么有些组合能关、有些不能关。它们回答不同问题，不能混成一个笼统的“鲁棒性测试”。

Test

All three tested growth conditions met the shutdown and counter criteria. Translation-flux FWHM was **5.48, 6.98, and 8.54 h** at doubling times of **40, 50, and 60 min**, respectively. These results showed that function was preserved across the tested conditions, while absolute timing changed with growth.

The joint parameter analysis revealed a more restricted operating region: **277 of 512 samples (54.1%)** satisfied the combined shutdown criteria. Thus, the architecture did not preserve acceptable performance across every sampled combination.

The transcription-supply scan helped explain this dependence. **124 of 225 combinations** passed. On the tested grid, the lowest passing effective sRNA-to-target transcription ratio was **1.39**, while the median minimum passing ratio across target-mRNA levels was **2.68**.

Within the joint sample, combinations with effective supply ratios between **4 and 8** had a **91.7%** pass fraction. Increasing supply further did not guarantee improvement, because some combinations suppressed the initial output peak too strongly.

同时保留正面结果和边界。三个生长条件都通过，说明这些条件下功能可以保留；但 54.1% 的联合通过比例说明性能仍依赖参数组合。4–8 是当前模型支持的优先筛选区间，不是普适阈值，更不是实验成功率。

Learn

The analyses supported retaining BM3R1–sRNA shutdown as a candidate architecture, with explicit operating constraints.

The main design lesson was that **effective sRNA supply must be evaluated relative to continuing target-mRNA production**. Merely enabling sRNA expression is insufficient: the module must balance adequate late suppression against preservation of the initial output peak.

This finding guides the next design decisions. Promoter strength and repression threshold should be evaluated as tuning variables, with effective supply ratios of **4–8** providing a starting range for construct screening. Increasing cassette dosage also increases BM3R1 operator load, so dosage changes require renewed checks of counter behavior. Absolute timing must be calibrated under the intended growth conditions.

The next stage is to select and characterize a specific sRNA–target construct and define its connection to the required counting state. The present model establishes conditional feasibility for a single-bit local shutdown module; experimental validation and integration with a complete target-count decoder remain outstanding.

最后落到“结果改变了什么设计决定”。保留这个架构，但不再把它描述成只要接上就能稳定关闭。后续要优先确定供给关系、标定培养条件下的时间，并检查对计数器的影响；完整计数状态译码和具体构建仍需要继续落实。